



Thursday, 08 Jan 2026

Knowledge Systems Working Group
Business & Primer Working Meeting

Robert Nilsson, Bill Hubbard



International Workshop 2026

Knowledge Systems Working Group

Robert Nilsson, Bill Hubbard



KSWG Meeting Schedule

- ☒ Sunday, 1 Feb 2026, - Business Meeting 90 min
- ☐ Monday, 2 Feb 2026 - Primer Review Meeting 120 min
- ☒ Tuesday, 3 Feb 2026 - Business & Joint WG Meeting 120 min

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

- ❑ Introductions (15 min)
- ❑ Knowledge Systems Overview, Scope, and Definitions (15 min)
- ❑ Knowledge Systems Taxonomy (15 min)
- ❑ Future State Vision / Role of AI (15 min)
- ❑ Working Group Cross-Connections (10 min)
- ❑ Primer Intro (10 min)
- ❑ Wrap up, closing comments, actions (10 min)

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

**Future State Vision / Role of AI
(15 min)**

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

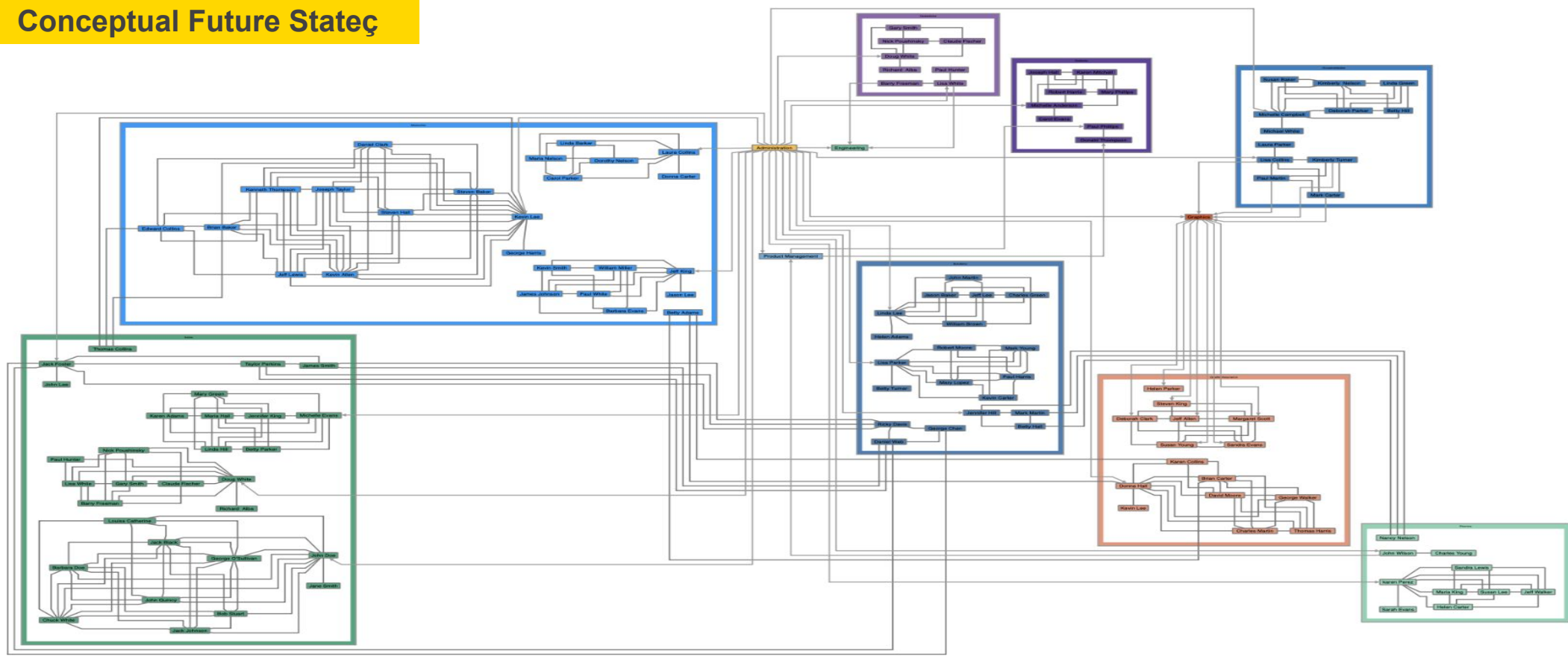
- ❑ How do we preserve meaning when knowledge crosses disciplinary boundaries?
- ❑ What knowledge do humans need to retain as AI systems become more capable—and how do we ensure that knowledge remains accessible?
- ❑ Where are the actual integration failures happening today—and are they semantic problems or something else?
- ❑ How do we handle the fundamental tension between formal precision and contextual flexibility?
- ❑ What happens when AI systems trained on different semantic frameworks need to collaborate—and who arbitrates conflicts?

KSWG Meeting Agenda

1 Feb 2026, Business Meeting, 15:30-17:00 PST

Future State Vision / Role of AI (15 min)

Conceptual Future State



KSWG Meeting Agenda

3 Feb 2026, Business Meeting & Joint WG Meeting,
10:00-12:00 PST

- ☐ Introductions (10 min)
- ☐ Knowledge Systems Overview, Scope, and Definitions (15 min)
- ☐ Systems Science Overview (15 min)
- ☐ Common fundamentals between Working Groups & future work (60 min)
- ☐ Revisit Knowledge Systems / System Science Taxonomy (10 min)
- ☐ Primer Edit (0 min)
- ☐ Wrap up, closing comments, actions (10 min)

KSWG Meeting Agenda

3 Feb 2026, Business Meeting & Joint WG Meeting,
10:00-12:00 PST

**Common fundamentals between
Working Groups & future work
(60 min)**

KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (10 min)

- ☐ When your Working Group's outputs need to be understood or used by another WG, what actually breaks?
- ☐ How much time does your WG spend "translating" concepts when collaborating with other groups?
- ☐ What terminology does your WG use that you know means something different in other contexts?
- ☐ If INCOSE had a shared vocabulary with agreed definitions across all Working Groups, would your WG actually use it?
- ☐ How can we discover additional findings with Traditional Analytics?

Bridging the Gap Between Engineering Disciplines

KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (10 min)

- ☐ What's the minimum viable semantic infrastructure that would provide value to your WG?
- ☐ How do you feel about AI systems interpreting and acting on your WG's knowledge?
- ☐ If your WG had to contribute to shared semantic infrastructure, what would you need in return?
- ☐ Who in your WG has the authority to commit to semantic standards—and would they?
- ☐ What's the cost of not having semantic interoperability over the next 5 years?

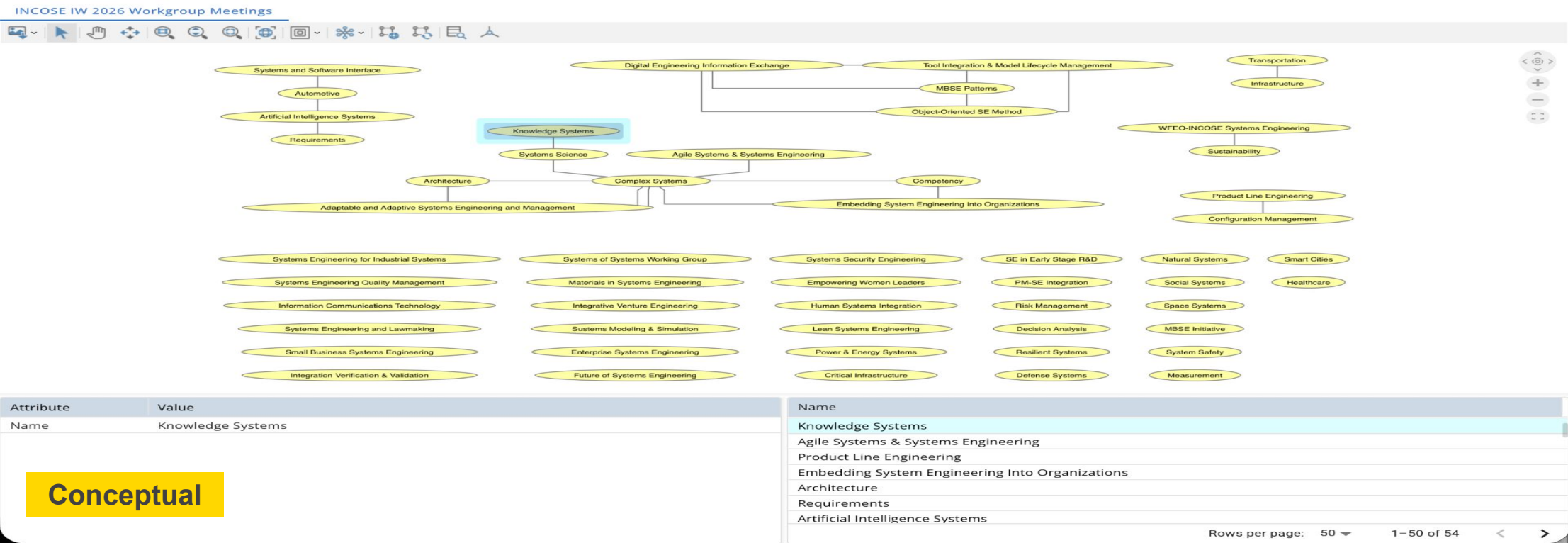
Bridging the Gap Between Engineering Disciplines

KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (10 min)



Conceptual

Conceptual Map to Bridge the Gap Between Engineering Disciplines

KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (20 min)

Common fundamentals between Working Groups & future work

**20 Min.
Group Exercise**

Create Relationships Between WGs

KSWG Meeting Agenda

1 Feb 2026, 15:30-17:00 PST

Foundational Interoperability (The Digital Core)

Common fundamentals between Working Groups & future work (20 min)

Ontology Mapping Exercise

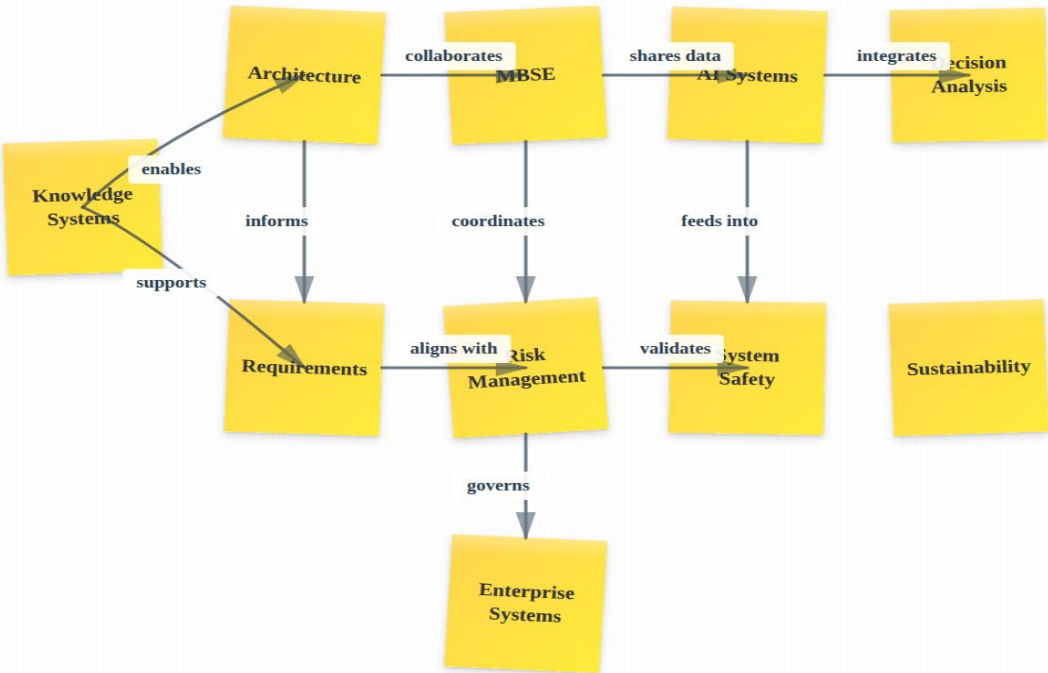
Working Groups Network

- 1 Identify Entities**
Place working groups on sticky notes (nodes in the network)
- 2 Draw Connections**
Connect groups that collaborate or interact (edges in the graph)
- 3 Label Relationships**
Describe the nature of each connection (relationship types)
- 4 Formalize Structure**
Transform the visual into a formal ontology model

Visual Key

-  Working Group (Node)
-  Relationship (Edge)

Network Diagram





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Today's Agenda



- ❑ Agenda items
 - Future meeting schedule.
 - International Workshop 2026 in Torrance California
 - **Working Group Collaborations**
 - Open Mic; Controlled Vocabulary, Semantics, Graph Analytics, Tools, AI4KS



INCOSE's International Workshop is the event of the year for systems engineers to contribute to their industry

Attendees spend four focused days working alongside fellow systems engineers to make a difference. Unlike INCOSE's annual **International Symposium** and other events, the IW is a paper, panel, or tutorial presentations. Systems Engineers at all levels are encouraged to engage in working sessions, and contribute their knowledge and experience to take the discipline forward.

What Working Groups do we want to collaborate with?

Sessions are encouraged to inclusively engage multiple working groups, diverse practitioners, and multiple competency levels as they complete meeting outcomes, satisfying the session's purpose. Attendees should experience a different yet complementary experience to regular working group meetings throughout the year.

IW 2026 Attendance

IW 2026 is open to **INCOSE** members, and non-members are encouraged to join **INCOSE** prior to the IW. For more information about membership, please visit <https://www.incose.org/incose-member-resources/join-incose> or email info@incose.net.

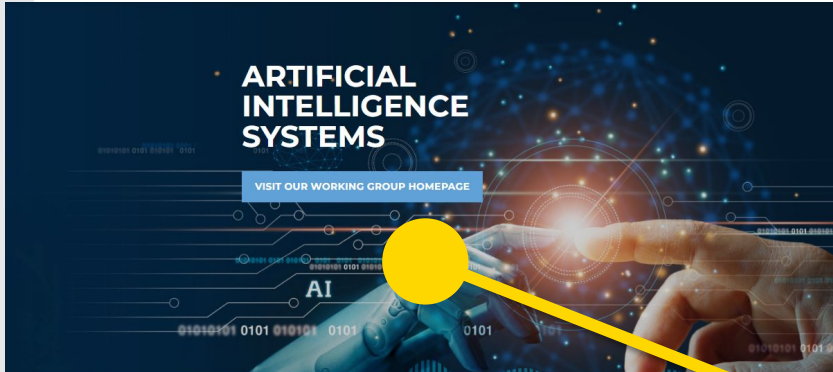
Working Group Collaborations



Artificial Intelligence Systems

Transformational

Ali K. Raz, Chair
Ramakrishnan "Ramki"
Raman, Co-Chair



COMPLEX SYSTEMS WORKING GROUP

VISIT OUR MEMBERS-ONLY INET PAGE



Complex Systems

Analytic Enablers

Dean Beale, Chair
Ali K. Raz, Co-Chair
Andrew C. "Andy"
Pickard, Co-Chair
Michael D. Watson, Co-Chair

Systems Science

Transformational

Javier Calvo-Amodio,
Chair
Harington Lee, Co-Chair
James N. Martin, Co-Chair



KNOWLEDGE SYSTEMS WORKING GROUP

Previously Knowledge Management



Knowledge Systems

Transformational

Robert Nilsson, Co-Chair
William Hubbard, Co-Chair



Today's Agenda



□ Agenda items

- Future meeting schedule.
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- **Open Mic; Controlled Vocabulary, Semantics, Graph Analytics, Tools, AI4KS**





INCOSE Knowledge Systems WG

